MATHEMATICS

# Applications and interpretation

Higher Level | Set A | Paper 1

## Challenge mock examination

Time allowed: 2 hours Maximum mark: 110

Candidate name: _____

Candidate number: _____

Date: _____

## Instructions

• A graphic display calculator and a clean Mathematics: applications and interpretation formula booklet are required.

• Answer all questions. Show working and explain mathematical reasoning in context.

• All questions are compulsory. There is no optional section.

• Give exact answers where appropriate. Otherwise give numerical answers to three significant figures, unless a question specifies a different accuracy.

• Keep unrounded values in subsequent calculations. State units where relevant.

• Diagrams are not necessarily drawn to scale. Use the stated data and conditions.

• The number in square brackets gives the maximum mark for a part.

Thriving Scholars original practice | 2026 challenge series

Question 1 | Question paper

## Question 1 [Maximum mark: 6]

A laboratory buys a scanner for 48 000 dollars. Its resale value falls by 16% each year. Maintenance costs 900 dollars in year 1 and rises by 18% each year. A replacement can be made at the end of any year $n$, where $1 \leq n \leq 12$. Define the average annual ownership cost as the purchase price minus resale value plus total maintenance, divided by $n$.

(a) Find the resale value after 5 years. [2]

(b) Find the total maintenance cost in the first 5 years. [2]

(c) Determine the replacement year that minimizes the average annual ownership cost over the permitted years. Give the minimum cost to the nearest dollar. [2]

## Question 2 [Maximum mark: 6]

A sensor is calibrated against known concentrations $x$ (mg $\mathrm{L}^{-1}$). Its measured signal is $y$ (mV).

| $x$ | 10 | 20 | 30 | 40 | 50 | 60 |
|---|---|---|---|---|---|---|
| $y$ | 18.3 | 30.5 | 42.1 | 58.4 | 66.7 | 82.6 |

Treat concentration as the controlled explanatory variable.

(a) Find the least-squares regression line of $y$ on $x$ and the Pearson correlation coefficient. [3]

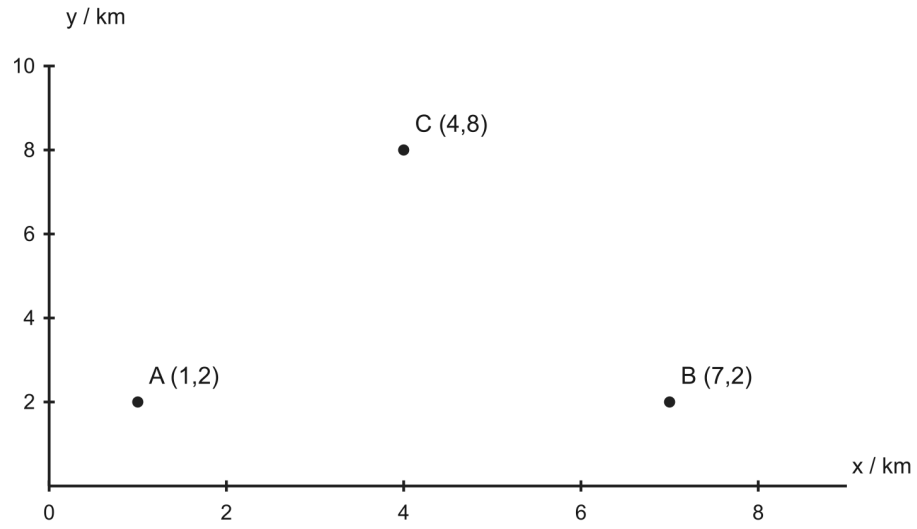
(b) Use your calibration line to estimate the concentration corresponding to a signal of 50 mV. Give the result to 5 decimal places. [1]

(c) Find the regression estimate of $x$ on $y$ when $y = 50$, to 5 decimal places. Explain why it need not equal the answer to part (b). [2]

Question 3 | Question paper

## Question 3 [Maximum mark: 6]

Three rescue stations are at $A(1, 2)$, $B(7, 2)$ and $C(4, 8)$ on a coordinate grid measured in kilometres. Each location is assigned to its nearest station. The diagram shows the sites, not the Voronoi boundaries.




(a) Write down the equation of the boundary equidistant from $A$ and $B$. [1]

(b) Find the coordinates of the Voronoi vertex at which all three cells meet. [3]

(c) A call is received at $(4, 5)$. State the station assigned to the call and calculate its straight-line distance from the call. [2]

## Question 4 [Maximum mark: 6]

A random sample of 150 trainees is classified by training programme and whether they pass an assessment.

| Programme | Pass | Fail |
|---|---|---|
| X | 36 | 14 |
| Y | 28 | 22 |
| Z | 40 | 10 |

Use a chi-squared test of independence at the 5% significance level.

(a) Find the expected number of passes in programme X under the null hypothesis, and state the degrees of freedom. [2]

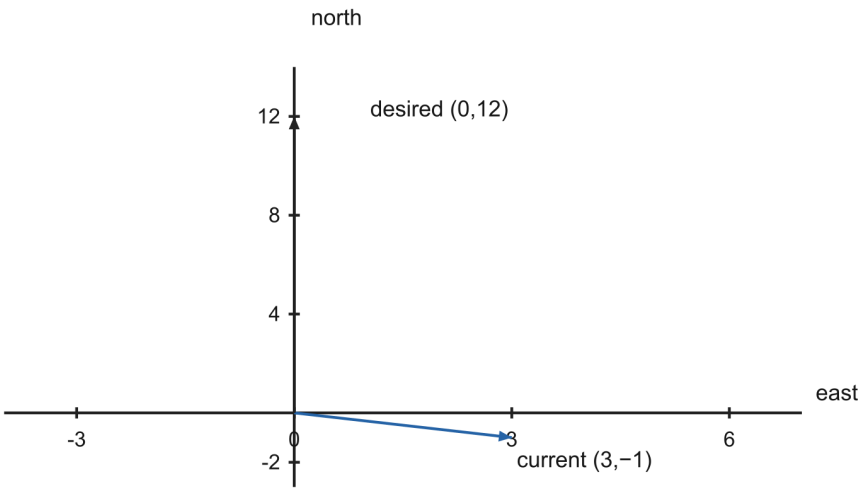
(b) Calculate the test statistic and the $p$-value. [2]

(c) State the conclusion in context. Explain why this test alone does not establish a causal effect of programme on passing. [2]

Question 5 | Question paper

## Question 5 [Maximum mark: 6]

A boat must travel due north with ground velocity $(0, 12)$ km $h^{-1}$. A uniform current has velocity $(3, -1)$ km $h^{-1}$. Vector components are east and north respectively. The ground velocity is the sum of the boat velocity relative to the water and the current.




(a) Find the required velocity of the boat relative to the water. [2]

(b) Find the required speed and bearing of the boat relative to the water. Give the bearing to the nearest degree. [3]

(c) The current changes to $(6, -2)$ while the boat keeps this same water-relative velocity. Find its eastward displacement after one hour. [1]

## Question 6 [Maximum mark: 7]

Independent component masses satisfy $X \sim N(500, 12^2)$ and $Y \sim N(200, 8^2)$, in grams. A package has mass $T = X + 2Y$. The same value of $Y$ is doubled; it is not the sum of two independent copies.

(a) State the distribution of $T$, including its mean and standard deviation. [3]

(b) Find $P(880 < T < 930)$. [2]

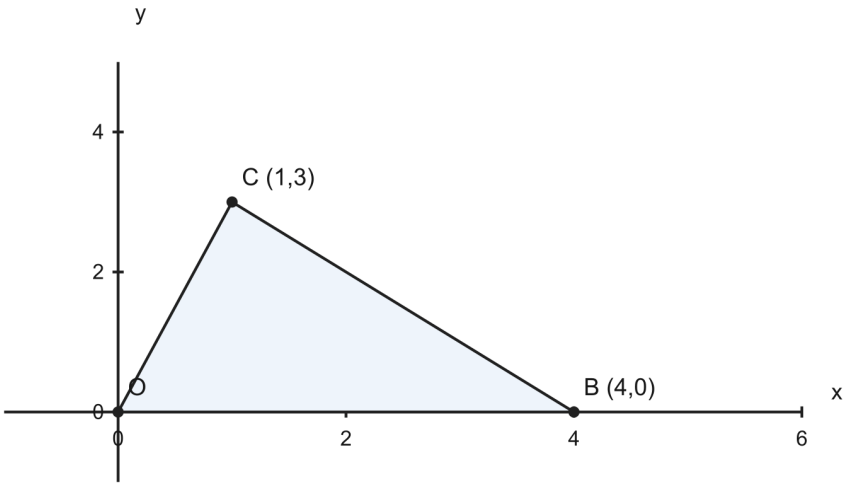
(c) Find the mass $c$ such that only 1% of packages exceed $c$. [2]

## Question 7 [Maximum mark: 7]

A linear transformation maps a column vector $\mathbf{x}$ to $M\mathbf{x}$, where

$$M = \begin{pmatrix} 1.2 & 0.3 \\ -0.4 & 0.8 \end{pmatrix}$$

. A triangular sign has vertices $O(0,0)$, $B(4,0)$ and $C(1,3)$; lengths are measured in metres.




(a) Find the images of $B$ and $C$. [2]

(b) Find the area of the transformed sign. [2]

(c) A transformed point is at $(3,2)$. Find its original position, showing an inverse-matrix method. [3]

Question 8 | Question paper

## Question 8 [Maximum mark: 7]

The number of registered users of a service is modelled by $P(t) = 1200/(1 + 7e^{-0.4t})$, where $t \geq 0$ is measured in days. Treat $P$ as continuous for this model.

(a) Find the initial number of users and the limiting number predicted by the model. [2]

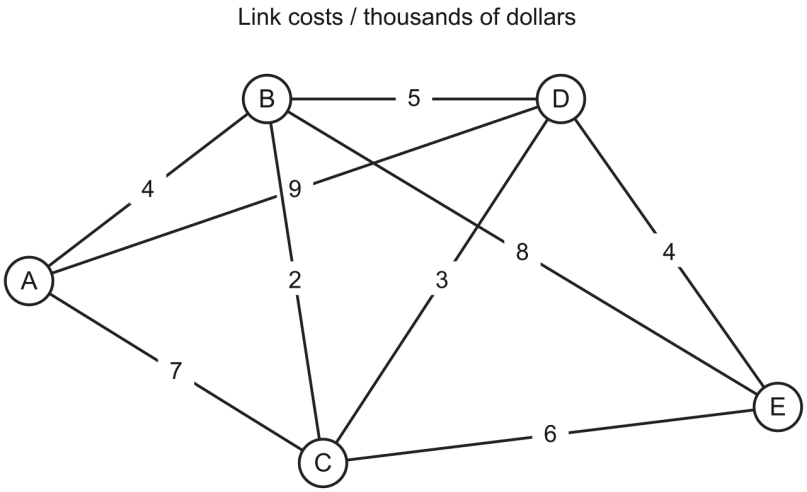
(b) Find the first time when the model predicts 900 users. [2]

(c) Find the maximum rate at which the number of users increases, and the time at which it occurs. [3]

Question 9 | Question paper

## Question 9 [Maximum mark: 7]

A conservation reserve needs fibre links connecting all five monitoring stations. The undirected graph shows the possible links and their construction costs in thousands of dollars. A connection is required between every pair of stations through the network, but a direct link is not required.



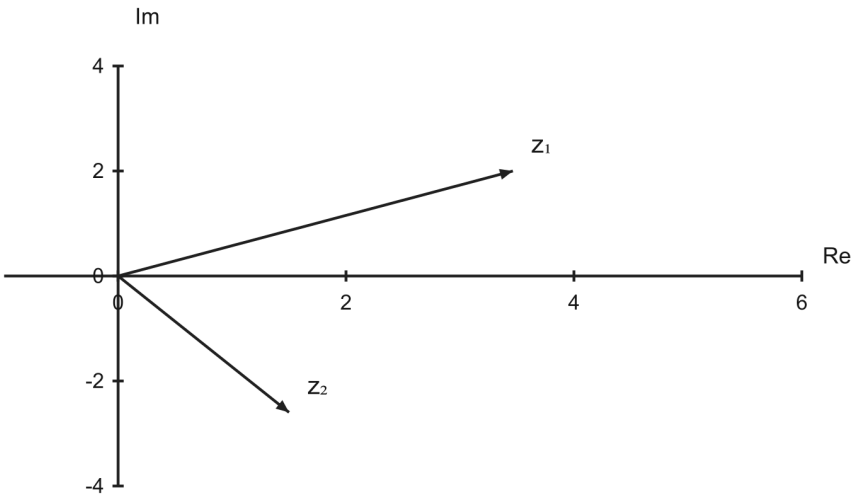

(a) Use Kruskal's algorithm to find a minimum spanning tree. State its edges in the order selected and its total cost. [4]

(b) Link $CD$ cannot now be built. Determine the revised minimum cost and list one optimal set of links. [3]

Question 10 | Question paper

## Question 10 [Maximum mark: 7]

Two oscillations of the same angular frequency are represented by $z_1 = 4e^{i\pi/6}$ and $z_2 = 3e^{-i\pi/3}$. Their resultant is $z = z_1 + z_2$. Arguments are in radians.




(a) Express $z$ in Cartesian form. [3]

(b) Find the amplitude $|z|$ and principal argument of $z$. [2]

(c) A third oscillation exactly cancels the resultant. Give its amplitude and an argument in $[0, 2\pi)$. [2]

## Question 11 [Maximum mark: 7]

On a randomly chosen operating day, a service desk is in weekday mode with probability 0.7 and weekend mode with probability 0.3. Conditional on the mode, the number of urgent calls in an hour is Poisson with mean 4 or 7 respectively. Counts in different hours and on different days are independent conditional on their modes.

(a) Find the probability of no urgent calls in the chosen hour. [2]

(b) Given that no calls occurred, find the probability that the desk was in weekday mode. [3]

(c) An hour in weekday mode and an independent hour in weekend mode are selected. Find the probability of at least 14 calls in total. [2]

Question 12 | Question paper

## Question 12 [Maximum mark: 7]

The same eight runners complete a fixed course before and after a training programme. The times, in seconds, are shown below. Assume the population of paired differences is normally distributed.

| Runner | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| Before | 82 | 77 | 91 | 85 | 88 | 79 | 94 | 83 |
| After | 78 | 76 | 85 | 81 | 86 | 75 | 88 | 80 |

Let $D$ = before time minus after time. Test for an improvement at the 5% level.

(a) Find the sample mean and unbiased sample standard deviation of the eight differences. [2]

(b) State the null and alternative hypotheses. [1]

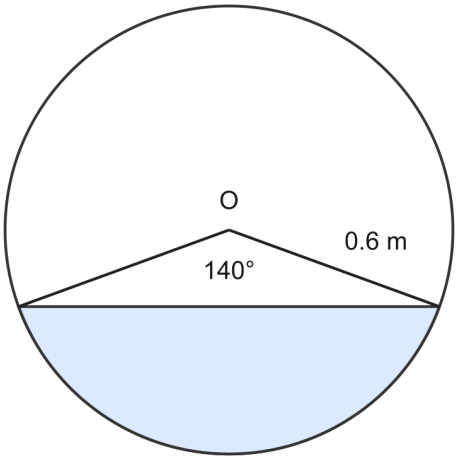
(c) Calculate the paired $t$ statistic and the one-tailed $p$-value. [2]

(d) State a conclusion in context and one additional sampling assumption needed for the inference. [2]

Question 13 | Question paper

## Question 13 [Maximum mark: 7]

A straight channel has a circular cross-section of radius 0.6 m. The water cross-section is the minor segment cut off by a chord subtending $140^\circ$ at the centre, as shaded. This cross-section is uniform along a channel length of 12 m.




Shaded minor segment is the water cross-section

(a) Calculate the cross-sectional area of the water. [3]

(b) Find the volume of water in the channel. [2]

(c) A floating inspection bar must span the water surface. Find its required length, ignoring any clearance. [2]

Question 14 | Question paper

## Question 14 [Maximum mark: 8]

A closed cylindrical tank must hold 4 $m^3$. Its circular top and base cost 12 dollars per $m^2$; its curved side costs 7 dollars per $m^2$. The radius $r$ is restricted to $0.45 \leq r \leq 1.4$ metres. Material thickness and waste are ignored.

(a) Show that its material cost is $C(r) = 24\pi r^2 + 56/r$. [3]

(b) Find the radius and height giving minimum cost, and justify that the minimum is attained. [3]

(c) A new rule limits the height to at most 2 m. Find the new optimal radius and minimum material cost. [2]

Question 15 | Question paper

## Question 15 [Maximum mark: 8]

A transport survey classifies a commuter each week as walking (W), cycling (C) or taking the bus (B). The regular transition matrix is

$$T = \begin{pmatrix} 0.8 & 0.2 & 0.1 \\ 0.15 & 0.6 & 0.2 \\ 0.05 & 0.2 & 0.7 \end{pmatrix}$$

. State vectors are columns in the order W,C,B, and $T_{ij}$ is the probability of moving from state $j$ to state $i$. A commuter walks in week 0.

(a) Find the probabilities of the three states in week 2. [3]

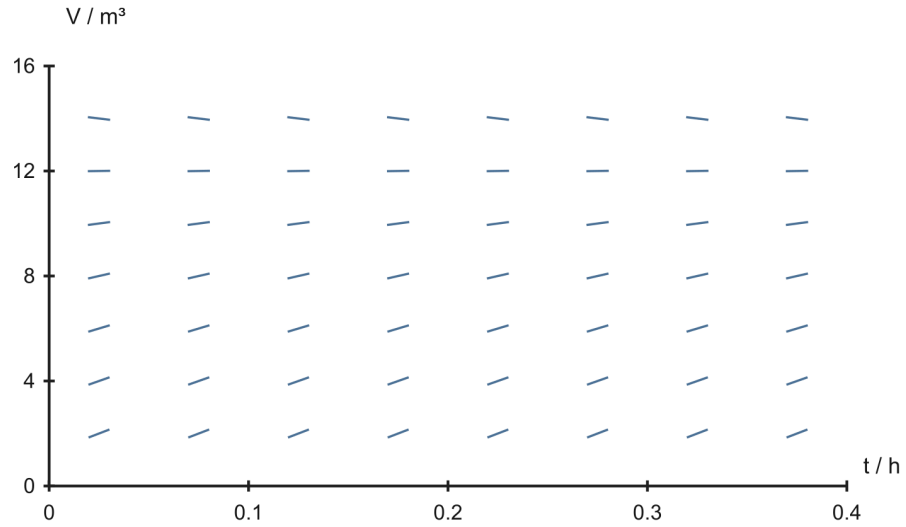
(b) Find the long-run proportion of weeks the commuter is expected to use each mode. [3]

(c) Given that the commuter cycles in week 2, find the probability that they took the bus in week 1. [2]

Question 16 | Question paper

## Question 16 [Maximum mark: 8]

A reservoir contains $V$ m$^3$ of water at time $t$ hours. It is modelled by $\frac{dV}{dt} = 30 - 0.2V^2$, with $V(0) = 4$. A slope field is shown. The model applies while $V \geq 0$.




(a) Find the physically relevant equilibrium volume and determine whether volumes just below it increase or decrease. [2]

(b) Use Euler's method with step size 0.1 hours to estimate $V(0.3)$. Show all three updates. [3]

(c) Calculate the one-step Euler estimate using step size 0.3. Explain, using concavity, whether tangent-line Euler estimates from points on the exact solution overestimate or underestimate the exact increasing solution. [3]

## End of question paper

MATHEMATICS

# Applications and interpretation

Higher Level | Set A | Paper 1

## Worked solution book

Complete this paper before consulting the worked solutions. Each part has method (M), accuracy (A) and reasoning (R) points. The number following a code is its mark value. Award only the points supported by the candidate's retained working.

Accept mathematically equivalent valid methods. Numerical answers may use technology, but should be supported by equations, test definitions, bounds, labelled graphs or other appropriate mathematical working. Retain unrounded intermediate values.

Use the portal to enter a whole-number self-assessed mark for each part. These are independent practice papers; the self-assessed report does not calculate an official IB grade.

Question 1 | Worked solutions

# Question 1 [Maximum mark: 6]

## (a) [2 marks]

**M1** Use $48000(0.84)^5$.

**A1** The resale value is $20074.173$ dollars, or $20\,074$ dollars to the nearest dollar.

## (b) [2 marks]

**M1** Use the geometric sum $900(1.18^5 - 1)/(1.18 - 1)$.

**A1** Total maintenance is $6438.7888$ dollars.

## (c) [2 marks]

**M1** Evaluate $C_n = [48000 - 48000(0.84)^n + 900(1.18^n - 1)/0.18]/n$ for integers $1 \le n \le 12$.

**A1** Replace at the end of year $11$; the minimum average is $6075$ dollars per year.

Question 2 | Worked solutions

# Question 2 [Maximum mark: 6]

## (a) [3 marks]

**M1** Apply linear regression to all six pairs.

**A1** $y = 1.2754286x + 5.1266667$.

**A1** $r = 0.99778088$.

## (b) [1 marks]

**A1** $x = (50 - 5.1266667)/1.2754286 = 35.18295 \text{ mg L}^{-1}$, using unrounded regression coefficients.

## (c) [2 marks]

**A1** The reverse regression is $x = 0.78057424y - 3.846578$, giving $35.18213 \text{ mg L}^{-1}$.

**R1** The regressions minimize residuals in different directions. They are inverses only for perfect linear correlation; here $|r| < 1$.

# Question 3 [Maximum mark: 6]

## (a) [1 marks]

**A1** $x = 4$.

## (b) [3 marks]

**M1** Equate squared distances to $A$ and $C$, giving $6x + 12y = 75$.

**M1** Solve this equation together with $x = 4$.

**A1** The vertex is $(4, 4.25)$.

## (c) [2 marks]

**M1** The distances to $A, B, C$ are $\sqrt{18}, \sqrt{18}, 3$ km.

**A1** Station $C$ is assigned; the distance is $3$ km.

Question 4 | Worked solutions

# Question 4 [Maximum mark: 6]

## (a) [2 marks]

**A1** $E_{X,\text{pass}} = 50(104)/150 = 34.666667.$

**A1** $df = (3-1)(2-1) = 2.$

## (b) [2 marks]

**M1** Use $\chi^2 = \sum (O-E)^2/E$ across all six cells.

**A1** $\chi^2 = 7.0234114$, $p = 0.029845963.$

## (c) [2 marks]

**R1** $p = 0.029845963 < 0.05$, so reject independence: there is evidence of an association between programme and passing.

**R1** An observational classification does not control confounding factors or randomly assign programmes, so association does not establish causation.

# Question 5 [Maximum mark: 6]

## (a) [2 marks]

**M1** Subtract the current from the required ground velocity.

**A1** The required velocity is $(-3, 13)$ km h$^{-1}$.

## (b) [3 marks]

**M1** $v = \sqrt{(-3)^2 + 13^2}$ and the heading is west of north.

**A1** $v = 13.341664$ km h$^{-1}$.

**A1** The bearing is $347^\circ$ ($347.00538^\circ$ before rounding).

## (c) [1 marks]

**A1** The new eastward ground speed is $-3 + 6 = 3$ km h$^{-1}$, so the displacement is $3$ km east.

# Question 6 [Maximum mark: 7]

## (a) [3 marks]

**M1** $E(T) = 500 + 2(200)$ and $\text{Var}(T) = 12^2 + 2^2(8^2)$.

**A1** $E(T) = 900$ g and $\sigma_T = 20$ g.

**A1** $T \sim N(900, 20^2)$.

## (b) [2 marks]

**M1** Use the normal distribution with the parameters in part (a).

**A1** The probability is $0.77453754$.

## (c) [2 marks]

**M1** $c = 900 + 20\Phi^{-1}(0.99)$.

**A1** $c = 946.52696$ g.

Question 7 | Worked solutions

# Question 7 [Maximum mark: 7]

## (a) [2 marks]

**A1** $B \mapsto (4.8, -1.6)$.

**A1** $C \mapsto (2.1, 2)$.

## (b) [2 marks]

**M1** The original area is $6\ \text{m}^2$ and the area scale factor is $|\det M| = 1.08$.

**A1** $6(1.08) = 6.48\ \text{m}^2$.

## (c) [3 marks]

**M1** $M^{-1} = \frac{1}{1.08}\begin{pmatrix} 0.8 & -0.3 \\ 0.4 & 1.2 \end{pmatrix}$

**M1** Multiply $M^{-1}(3, 2)^T$.

**A1** The original point is $(1.6666667, 3.3333333)$.

# Question 8 [Maximum mark: 7]

## (a) [2 marks]

**A1** $P(0) = 150$.

**A1** The limiting number is 1200 users.

## (b) [2 marks]

**M1** Solve $1200/(1 + 7e^{-0.4t}) = 900$.

**A1** $t = 7.6113061$ days.

## (c) [3 marks]

**M1** $P\prime(t) = 0.4P(1 - P/1200)$, maximized when $P = 600$.

**A1** The maximum rate is 120 users per day.

**A1** This occurs at $t = \ln(7)/0.4 = 4.8647754$ days.

# Question 9 [Maximum mark: 7]

## (a) [4 marks]

**M1** Consider edges in increasing order and reject an edge if it creates a cycle.

**A2** A valid order is $BC, CD, AB, DE$ (the two edges of cost 4 may be interchanged).

**A1** The total cost is 13 thousand dollars.

## (b) [3 marks]

**M1** Repeat the cycle-free selection with $CD$ excluded.

**A1** Use $BC, AB, DE, BD$.

**A1** The revised cost is 15 thousand dollars.

# Question 10 [Maximum mark: 7]

## (a) [3 marks]

**M1** Resolve each phasor into real and imaginary components.

**A1** $\operatorname{Re} z = 2\sqrt{3} + 3/2$.

**A1** $z = 4.9641016 - 0.59807621i$.

## (b) [2 marks]

**A1** $|z| = 5$.

**A1** $\arg z = -0.11990233$ rad.

## (c) [2 marks]

**M1** The cancelling phasor is $-z$, so add $\pi$ to the argument.

**A1** Amplitude $5$ and argument $3.0216903$ rad.

# Question 11 [Maximum mark: 7]

## (a) [2 marks]

**M1** Use the total-probability sum $0.7e^{-4} + 0.3e^{-7}$.

**A1** The probability is $0.013094512$.

## (b) [3 marks]

**M1** Use Bayes' theorem with numerator $0.7e^{-4}$.

**M1** Divide by $0.013094512$.

**A1** The conditional probability is $0.97910845$.

## (c) [2 marks]

**M1** The sum is Poisson with mean $4 + 7 = 11$.

**A1** $P(X \geq 14) = 1 - P(X \leq 13) = 0.21870883$.

# Question 12 [Maximum mark: 7]

## (a) [2 marks]

**A1** $\bar{d} = 3.75$ s.

**A1** $s_D = 1.7525492$ s.

## (b) [1 marks]

**A1** $H_0 : \mu_D = 0, H_1 : \mu_D > 0.$

## (c) [2 marks]

**M1** $t = \bar{d}/(s_D/\sqrt{8})$ with $df = 7$.

**A1** $t = 6.0520994, p = 0.00025744373.$

## (d) [2 marks]

**R1** $p = 0.00025744373 < 0.05$, so there is evidence that mean running time decreases.

**R1** The pairs should be a random/representative sample and differences from different runners should be independent. Do not assume the before and after measurements on the same runner are independent.

Question 13 | Worked solutions

# Question 13 [Maximum mark: 7]

## (a) [3 marks]

**M1** Convert $140^\circ$ to $\theta = 7\pi/9$ radians.

**M1** Subtract the isosceles triangle from the sector: $A = \frac{1}{2}r^2(\theta - \sin\theta)$.

**A1** $A = 0.3241212\ \text{m}^2$.

## (b) [2 marks]

**M1** Multiply the cross-sectional area by 12.

**A1** $V = 3.8894544\ \text{m}^3$.

## (c) [2 marks]

**M1** The chord length is $2r\sin(\theta/2)$.

**A1** The length is $1.1276311\ \text{m}$.

Question 14 | Worked solutions

# Question 14 [Maximum mark: 8]

## (a) [3 marks]

**M1** $h = 4/(\pi r^2)$ from the volume constraint.

**M1** Top/base cost $12(2\pi r^2)$ and side cost $7(2\pi r h)$.

**A1** Hence $C = 24\pi r^2 + 56/r$.

## (b) [3 marks]

**M1** $C\prime(r) = 48\pi r - 56/r^2 = 0$.

**A1** $r = 0.71878495$ m, $h = 2.4644055$ m.

**R1** $C''(r) = 48\pi + 112/r^3 > 0$, and this stationary point lies inside the permitted interval.

## (c) [2 marks]

**M1** $h \leq 2$ gives $r \geq \sqrt{2/\pi}$, above the unconstrained optimum, where cost is increasing.

**A1** Choose $r = 0.79788456$ m, $h = 2$ m; cost $118.18559$ dollars.

# Question 15 [Maximum mark: 8]

## (a) [3 marks]

**M1** Use $T^2(1,0,0)^T$.

**A2** The state vector is

$$\begin{pmatrix}0.675\\0.22\\0.105\end{pmatrix}$$

## (b) [3 marks]

**M1** Solve $T\boldsymbol{\pi} = \boldsymbol{\pi}$ with $\pi_W + \pi_C + \pi_B = 1$.

**A2** $$\boldsymbol{\pi} = \begin{pmatrix}0.43243243\\0.2972973\\0.27027027\end{pmatrix}$$

## (c) [2 marks]

**M1** The path W to B to C has probability $0.05(0.2) = 0.01$; divide by $P(C_2) = 0.22$.

**A1** The conditional probability is $0.045454545$.

Question 16 | Worked solutions

# Question 16 [Maximum mark: 8]

## (a) [2 marks]

**A1** The non-negative equilibrium is $V = \sqrt{150} = 12.247449\ \text{m}^3$.

**R1** For $0 \leq V < \sqrt{150}$, $30 - 0.2V^2 > 0$, so volume increases.

## (b) [3 marks]

**M1** $V_{k+1} = V_k + 0.1(30 - 0.2V_k^2)$.

**A1** $V_1 = 6.68$ and $V_2 = 8.787552$.

**A1** $V_3 = 10.243131\ \text{m}^3$.

## (c) [3 marks]

**A1** The one-step estimate is $4 + 0.3(30 - 0.2 \times 4^2) = 12.04\ \text{m}^3$.

**M1** $V'' = -0.4VV\prime < 0$ while $0 < V < \sqrt{150}$.

**R1** The solution is concave down, so its forward tangent lines lie above the solution; tangent-line Euler estimates overestimate. The smaller-step result is closer here, but that observation alone is not a general proof.

# End of worked solution book